

# Does a Music Transformer Know What Key It Is In? Causal Interventions on an Emergent Tonal World Model

Anonymous ICASSP 2027 submission

**Abstract**—Music generation models produce convincingly tonal music. Whether they chain surface statistics or maintain an internal representation of musical key that generation *causally uses* has not been tested. We test it directly, under a pre-registered protocol with frozen decision rules, on Transformers trained on symbolic music whose vocabulary holds absolute pitch and time only—no key, chord, or degree tokens. Replacing the key component of the residual stream at a single middle layer makes the model continue in the injected key at  $5.0\times$  the rate of rank-matched random edits (guarded TKR 0.378 vs. 0.075; significant for 12/12 targets after Holm correction, replicated across seeds), inside a musicality budget frozen before any edit ran, reaching 58% of the ceiling set by transposing the prompt itself. Two controls say what this requires. Across a  $170\times$  capacity range next-token accuracy is flat, yet at 0.5M parameters the key is not decodable beyond the input surface at all and the same edit moves the key *more* often than at 26M while wrecking the music 90% of the time: capacity does not buy the key, it buys a key that can be moved without breaking everything else. Finally we leave our own models: on real Bach chorales a *public* checkpoint trained on real music—architecturally identical to one of ours—carries a key state that beats the pitch surface where ours does not, and editing that state redirects the music it composes at  $5.7\times$  the matched control (11/12 targets, 100% within the frozen musicality budget). It also resists: the keys the edit cannot reach are the keys real music rarely uses. Editing a music Transformer’s internal key state changes what key it composes in—in a model people actually use, not only in ours.

**Index Terms**—music generation, world models, interpretability, causal intervention, tonality

## I. INTRODUCTION

Systems that compose music write convincingly tonal pieces: they settle into a home key, build tension away from it, and return. How they do so is unexamined. Analyses report that musical attributes can be *read out* of a model’s activations by trained classifiers [6]. Reading is not using: a signal can be decodable from a system’s state without the system consulting it when it acts [10]. No study has tested whether a music model *uses* an internal key representation—whether intervening on it, with the input untouched, changes the key of the music it goes on to generate.

Sequence models can build causal models of the latent state behind their data. A model trained only to predict

Othello moves tracks the board those moves imply, and editing that internal board changes which moves it treats as legal [1], [2]; analogous causal evidence exists for chess, mazes, and RL environments [3], [4], [5]. In music, evidence has stopped at readability [6], and steering work controls attributes without testing a state variable [7]. We occupy the empty cell: causal, state-level evidence in music.

Key is an unusually good state variable for this. It is defined *externally* to the model—by centuries of theory and by psychological measurement [9]—so the probed state is not an artifact of our labeling; it must be *computed* by integrating pitch evidence over time, unlike metric position, which time tokens hand over; and diatonic membership gives a natural analogue of Othello’s legal-move rate, so success is measurable without a learned judge.

Contributions: (1) a controlled testbed—leak-free tokenizer, functional-harmony corpus with exact per-token key labels, pre-registered protocol—released as a benchmark; (2) a causal result: single-layer subspace edits redirect the composed key for 12/12 targets inside a frozen musicality budget; (3) two controls that say what a world model requires: a capacity sweep separating the state’s presence, its legibility, and its manipulability; and the same probe AND the same intervention carried to a public model trained on real music, which reads its key state and uses it.

## II. METHOD

### A. Data, models, protocol

Ground truth must be exact, not estimated, so training data are synthetic. **D-SYN**: a functional-harmony state machine ( $T \rightarrow S \rightarrow D \rightarrow T$  with substitutions, SATB voicing, optional diatonic melody) generates 200k/10k/10k train/val/test pieces of 278–508 tokens with 0–3 modulations each (pivot, direct, or sequential; fifths-distance stratified), giving every token a ground-truth key label  $\kappa(t)$  (tonic  $\times$  mode, 24 classes). **Tokenizer**: the vocabulary’s 124 types are BAR, POS, absolute-MIDI PITCH and DUR (121 types) together with PAD/BOS/EOS; nothing names a key, chord or degree, and a unit test enforces this, so key must be computed from the notes. **Models**: M-CTRL is a GPT-2 style decoder (8 layers, 8 heads,  $d=512$ , ctx 512,  $\sim 25$ M params), trained under two regimes—with (R-Aug) and without (R-NoAug) uniform 12-fold transposition augmentation— $\times 3$  seeds. M-REF shares the

architecture but a disjoint seed *and* data split, and serves only as the musicality guard, avoiding circular evaluation. All models pass a pre-frozen quality gate (val. next-token top-1 0.878–0.881 vs. a 0.434 constant-predictor rate). Decision rules, thresholds and controls were frozen before any experiment ran; deviations are logged and negative results reported.

### B. Probing (Phase A)

Linear probes map residual-stream activations  $h_\ell(t)$  to  $\kappa(t)$  with sequence-level splits, against three control families: **C1**, a control task [10] permuting key identities per sequence; **C2**, an untrained model; **C3**, input-side baselines at the same positions (pitch-class-histogram logistic regression and Krumhansl–Schmuckler (KS) template matching, windows  $W \in \{8, 16, 32, 64\}$ ). **DR-H1**: the probe supports H1 iff its selectivity-corrected  $F_1$  exceeds the best C3 baseline with a sequence-level BCa 95% CI excluding zero in some layer.

The pre-registered hypotheses are H1 (key is decodable above the input surface), H2 (the representation is transposition-equivariant with cyclic-group structure) with sub-claim H2b (augmentation strengthens equivariance—hence no “H2a”: H2 is the parent), H3 (a targeted subspace edit moves the composed key, beating a rank- and norm-matched random edit inside the guard), H4 (after a *one-shot* edit the state persists, is reasserted, or transitions via a pivot), and H5 (causally effective layers are localized). H4 is Phase C: the pre-registration assigned it to the journal version before any run, so it is out of scope here by design rather than for space, and it has not been run.

### C. Intervention (Phase B)

From bar 9 ( $t^*$ ) of a key-stable 8-bar prompt, a sustained edit replaces the key component at one layer at *every* token position from  $t^*$  on (not only where a pitch is chosen—we write to the whole stream but probe one position in it), at every generation step,

$$h \leftarrow h - P_V h + P_V \mu_{\kappa^*}, \quad (1)$$

with  $V$  an orthonormal key subspace,  $P_V = VV^\top$ , and  $\mu_{\kappa^*}$  the class-conditional mean of the donor key  $\kappa^*$ . Subspaces: **V-PROBE** (probe row space, rank 24), **V-MEAN** (class-mean span), **V-DAS** (interchange-trained, ranks 8/24). Controls: **K1** rank-matched random subspace (H3 as pre-registered asks for rank *and* norm matching; we do not rescale, and report the measured magnitudes instead— $\|\Delta\|$  32.5 vs. 28.7, ratio 1.13, so the effect is not a magnitude artifact); **K2** sham edit, required bit-identical to the clean run (passed, 100/100); **K3** another layer’s subspace; **K4** the behavioral ceiling—transposing the prompt itself. Continuations (16 bars; 12 major targets  $\times$  100 prompts  $\times$  8 layers; sampling noise paired with a clean twin) are scored by KS key estimation (TKR strict: the continuation’s estimated key equals  $\kappa^*$ ; all headline

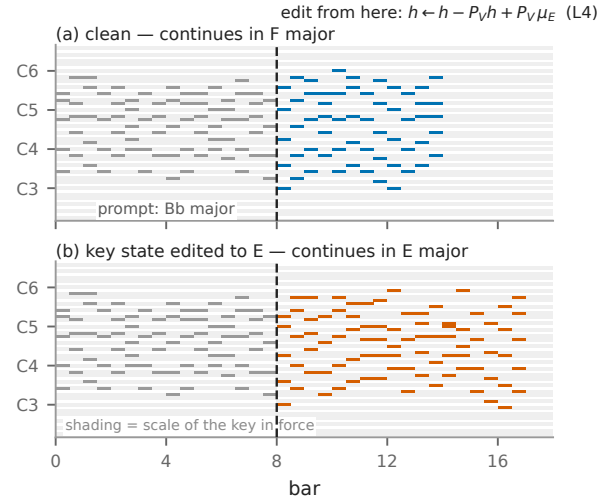


Fig. 1. The claim in one example (real model output, fixed sampling seed). The same 8-bar Bb-major prompt, continued (a) clean and (b) with the key component of layer-4 activations replaced by the E-major class mean from bar 9 on. **The input does not change**; the continuation’s estimated key follows the injected internal state. Shading marks the scale of the key in force.

numbers are strict) and by IKR (fraction of continuation pitches diatonic to a key). The pre-registered *tolerant* variant, used only to bracket estimator confusions, counts fifths distance  $\leq 1$  with the same mode, the relative, or the parallel key. The musicality budget  $\delta_{\text{PPL}}=0.613$ —the 90th percentile of the M-REF perplexity rise across 7,989 *natural* modulations in validation data—was fixed *before* any edit ran; edits exceeding it do not count as successes. **DR-H3** requires  $\text{TKR}(\text{edit}) > \text{TKR}(\text{K1})$  with Holm-corrected Wilcoxon  $p < .05$  in  $\geq 8/12$  targets at the best in-budget condition.

## III. RESULTS

Intervention numbers are from the R-Aug seed-0 sweep; probing holds for all six M-CTRL models, and the seed-1 sweep replicates every Phase-B verdict, though seed 1 peaks at L2 (TKR 0.321 vs. K1 0.056): the localization replicates, its depth does not.

### A. The key is decodable beyond the input surface

The key is *computed*, not transcribed: at L4 the linear probe reaches macro- $F_1$  0.927 against 0.790 for the best input baseline (PC-hist LR,  $W=64$ ) and a 0.032 selectivity floor, so the corrected margin is +0.105 (BCa 95% CI [0.091, 0.114]). The margin excludes zero in layers 1–7 of every trained model. It fails only at L0, where the margin is significantly *negative* (−0.20, CI [−0.211, −0.182]): at the embedding the notes themselves beat the network’s own read of them, exactly as they should. DR-H1 is supported 6/6. An untrained model scores 0.243. The internal state is worth most where the surface is worst: in the top quartile of KS-entropy ambiguity, probe accuracy is 0.936 vs. 0.795 for the baseline.

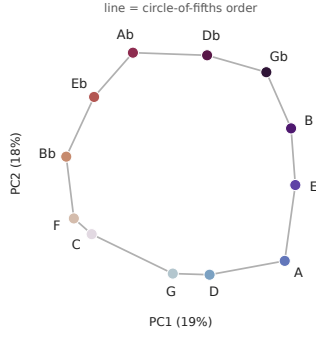


Fig. 2. The twelve major-tonic probe directions (L4, first two PCs) self-organize into a circle of fifths. No key, chord, or degree token exists in the vocabulary.

### B. Fifths geometry; equivariant but not a group

The representation carries music-theoretic structure. Projected to two principal components, the twelve major-tonic probe directions order themselves as C–G–D–A–E–B–F♯–... (Fig. 2): circle-of-fifths neighbour agreement 1.00 vs. 0.00 for the chromatic ordering, at every probed layer (exploratory). Transposition acts as a generalizing linear map—Procrustes  $R_k$  fit on half the sequences aligns held-out activations with 0.05–0.11 relative error—but the family is not cyclic ( $\varepsilon_{\text{cyc}} \approx 0.98$ , vs.  $\approx 1.41$  for random orthogonal maps), and our pre-registered comparison is *negative*: augmentation does not improve cyclicity (3 seeds per regime; 0.983 vs. 0.981; one-sided  $p = 0.83$ ), so DR-H2b is not supported.

### C. Editing the key state changes the composed key

At the best in-budget condition (V-PROBE, L4),  $\text{TKR}(\text{edit}) - \text{TKR}(\text{K1}) > 0$  holds for 12/12 targets (Holm-corrected  $p < 10^{-4}$ ; rank-biserial  $r = 0.64$ –1.00) against a pre-registered bar of  $\geq 8/12$ : **DR-H3 is supported** (Fig. 3). Pitch content triangulates the same conclusion without the estimator:  $\text{IKR}_{\kappa^*} = 0.937$  vs.  $\text{IKR}_{\kappa_{\text{src}}} = 0.646$  after editing. Success falls gently with fifths distance (0.395 at distance 1 to 0.280 at the tritone), while random edits collapse to zero once the identity target is excluded. The edit reaches 58% of the behavioral ceiling (K4): the state is readable in a rank-24 subspace of one layer, yet a single-layer edit realizes only part of the behavioral effect—an *actionability gap*. (Guarded edit over unguarded K4, whose perplexity twin is untransposed and so scores transposition, not damage; guarding both sides gives 62%, neither 66%—we report the smallest.) Two subsidiary findings: the *optimized* subspace (V-DAS, 0.213) underperforms the probe’s readout directions, and V-MEAN edits destroy quality at deep layers (guard pass  $\leq 5\%$ )—the guard is load-bearing, as Sec. III-E makes decisive.

### D. Readability is diffuse; causal efficacy is sharp

The edit-minus-control effect is single-peaked across layers (0.03, 0.04, 0.19, 0.26, **0.30**, 0.26, 0.23, 0.23 for L0–L7);

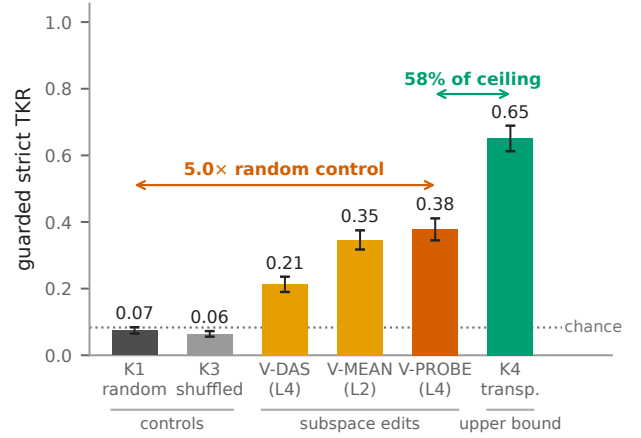


Fig. 3. Guarded strict TKR by condition: the continuation lands exactly on the injected key *and* stays inside the frozen musicality budget. Chance = 1/12. Bars: prompt-level bootstrap 95% CI.

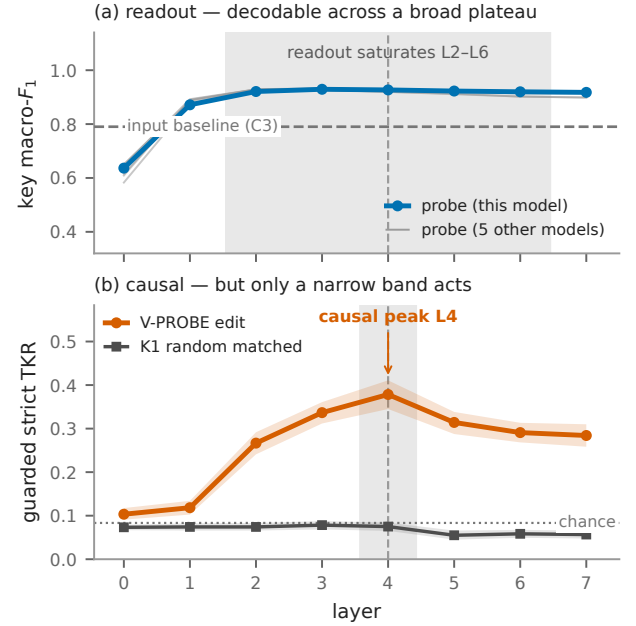


Fig. 4. (a) Probe macro- $F_1$  per layer, all six M-CTRL models (thin lines; R-Aug seed 0 highlighted): the key is legible across a broad plateau. (b) Causal effect: guarded TKR of the V-PROBE edit vs. the K1 matched random control (bands: prompt-level bootstrap 95% CI). Only a narrow band around L4 moves the output.

the best-vs. -median bootstrap CI [0.038, 0.109] excludes zero (DR-H5 supported). The two profiles in Fig. 4 are *not* the same shape, and that is the finding. The probe saturates across L2–L6 (macro- $F_1$  0.920–0.929, all within 0.01 of maximum): the key is legible from most of the network. Yet a single-layer edit moves the output only near L4, and barely at all at L0–L1. *Where a state is legible is a weak guide to where it is used*—a caution for work that localizes a computation by probing accuracy alone.

### E. What a world model actually requires

Training the same data and protocol at four capacities (L2/d128, L4/d256, L8/d512, L12/d768; 0.5M, 3.4M, 26M, 86M parameters counted from the checkpoints) dissociates three things the term “world model” conflates (Table I, top). *Behaviour* is flat: next-token top-1 is 0.868–0.879 across this  $170\times$  range. *Legibility* is not: at 0.5M the probe peaks *below* the input baseline and DR-H1 is unsupported in both seeds—a behaviourally equivalent model with no key state, the surface-statistics account made flesh, which the pre-registered test correctly refuses to credit. *Manipulability* is a third thing again: at 0.5M the edit moves the key *more* often than at 26M in raw terms (0.55 vs. 0.43) and survives the musicality guard only 10% of the time (vs. 80%)—with two layers, “editing the key subspace” is a blunt overwrite, not the movement of a separable state. Reporting guarded TKR alone hides this; reporting raw TKR alone credits a surface model with a world model. The guard is what tells them apart. Nor does manipulability simply keep improving: it peaks near 26M and falls to 0.18 at 86M, and a full twelve-layer sweep confirms this is not a layer-choice artifact (the 86M causal peak, L6 at 0.206, is indistinguishable from the L3 we had used, 0.202).

### F. A public model trained on real music: it reads, and it uses

Everything above is our own model on our own data. We close by leaving both. On 300 Bach chorales whose *local* key at every note comes from published Roman-numeral analyses,<sup>1</sup> our 86M model’s key subspace transfers ( $F_1$  0.534 vs. a 0.035 selectivity floor) but does *not* beat the pitch surface: the corrected margin, +0.034 (CI [−0.021, 0.103]), does not exclude zero.

The Anticipatory Music Transformer [8] (Apache-2.0; Lakh MIDI, MetaMIDI, FMA, 450k commercial records) is, at its **small** size, *architecturally identical* to that model—12 layers,  $d=768$ , 85.1M vs. 85.2M non-embedding parameters (its 128M total only reflects a 55k vocabulary against our 124)—and its (arrival-time, duration, note) vocabulary is leak-free exactly as ours is. Probed on the same chorales, at the same positions, with the same controls and rule (the two C3 baselines land at 0.451 and 0.454, confirming the setups match), it reaches  $F_1$  0.690 with a corrected margin of +0.196 (CI [0.132, 0.260]): **DR-H1 supported**. With architecture, corpus, protocol and rule fixed, *the training data is the single free variable*, and it decides whether a key state is worth more than the notes themselves.

**And that state is used.** We ran the same intervention on it: the probe’s row space as the subspace, a *different* public checkpoint as the guard’s reference (medium refereeing small—different weights, not circular;

<sup>1</sup>Scores: **bach-370-chorales** (CC BY-NC-SA); analyses: *When in Rome* (CC BY-SA). Neither is redistributed.

TABLE I

WHAT A WORLD MODEL REQUIRES. **Top:** SAME DATA AND PROTOCOL, FOUR CAPACITIES — BEHAVIOUR IS FLAT WHILE LEGIBILITY AND MANIPULABILITY ARE NOT. **Bottom:** SAME ARCHITECTURE (12 LAYERS,  $d=768$ ), SAME CHORALES, SAME PROTOCOL; ONLY THE TRAINING DATA DIFFERS. THE PUBLIC MODEL BOTH READS THE KEY AND USES IT.

| Model                                                   | top-1 | probe         | DR-H1      | guard |
|---------------------------------------------------------|-------|---------------|------------|-------|
| <i>capacity (D-SYN, our models)</i>                     |       |               |            |       |
| 0.5M                                                    | 0.870 | below C3      | no         | 10%   |
| 3.4M                                                    | 0.878 | +0.05         | yes        | 60%   |
| 26M                                                     | 0.879 | +0.10         | yes        | 80%   |
| 86M                                                     | 0.879 | +0.10         | yes        | 71%   |
| <i>real music, matched architecture (Bach chorales)</i> |       |               |            |       |
| ours, synthetic-trained                                 | —     | +0.034        | <b>no</b>  | —     |
| public, real-trained [8]                                | —     | <b>+0.196</b> | <b>yes</b> | —     |

$\delta_{\text{PPL}}=0.849$  nats frozen from 1,180 natural modulations in the chorales), the edit layer searched on 20 chorale prefixes and judged on 60 it never saw, sham edit bit-identical (20/20). **DR-H3 is supported:** at L8 the edit lands the continuation on the injected key at  $5.7\times$  the matched random control (0.365 vs. 0.064), significant in 11/12 targets, with IKR crossing (0.824 to the injected key vs. 0.785 to the prompt’s) and—unlike our own model’s 87%—100% **of edits inside the frozen musicality budget**. A key state learned from real music is *more* cleanly separable, not less.

Two things only a real model could show. The causal peak (L8) is again not the probe peak (L10), so Sec. III-C’s dissociation is not an artifact of synthetic data. And the model *resists*: the one target it cannot reach is F $\sharp$  major (TKR 0.000), and across all twelve targets success is ordered almost perfectly by how common that tonic is in the chorale corpus, counted over its note-weighted local-key labels (Spearman +0.91,  $n=12$ , one-sided permutation  $p < 10^{-4}$ ; +0.75 if keys are counted once per chorale instead). A model trained on real music can be steered into the keys that music favours and resists the rest—a prior our uniformly-keyed synthetic corpus could not have revealed. The chorale distribution is a proxy for tonal key frequency, not a measurement of this model’s own training corpus (Lakh MIDI and MetaMIDI, not chorales), so the prior’s *source* is not identified here—only its existence. One caution earned the hard way: the probe must be read where the model is about to *choose* a pitch. Read at the note instead, the same model on the same chorales decays with depth ( $F_1$  0.545 at L0 to 0.093 at L10) rather than peaking deep, and the margin falls to +0.042 (CI [−0.034, 0.092]): DR-H1 would be *unsupported*. One position choice, and the verdict inverts.

## IV. DISCUSSION AND LIMITATIONS

The evidence supports the world-model reading within this training condition: a Transformer trained on notes

alone maintains a mid-layer key variable with music-theoretic geometry, and generation consults it. The gap to the behavioral ceiling and the failure of edits at L0–L1 point to a distributed, redundantly re-estimated state rather than a single canonical register—a contrast with Othello-GPT’s near-complete editability that we attribute to music’s continual re-grounding of tonality in the preceding notes. K3 agrees: it is a negative control only at L4, where it borrows L0’s failed subspace and does nothing (0.064, at or below K1), but subspaces borrowed from L5–L7 and injected at L1–L3 reach 0.27–0.39, so the subspace is largely *shared* across the middle stack and its layer-pooled mean (0.160) is not a “partial effect”. The full version resolves K3 by layer.

**Bounds.** *Mode*: the pre-registration fixed the injected mode to major; minor targets were designated secondary and have not been run, so every causal claim here concerns major targets. *V-DAS below V-PROBE*: the optimized subspace may be undertrained relative to its search space; we report it as observed. *Estimator*: the KS judge confuses fifths-neighbours, which strict/tolerant reporting brackets and IKR triangulates. *Capacity*: the sweep varies depth and width together, locating a threshold, not its cause. *Selection*: the best (method, layer) is chosen on the data the Wilcoxon then tests, as pre-registered. *Real models*: the causal result holds for one public checkpoint at one size. The journal version carries the ambiguity strata, the specificity matrix, the fifths-distance curve, per-seed figures, and whether edited continuations modulate *musically*. Training data are synthetic; evaluation corpora are used under their stated licenses.

## V. CONCLUSION

A music Transformer tracks musical key as an internal state variable that generation consults—in our models, and in one people actually use. Capacity supplies not the key but a key that can be moved without breaking the music; the training data, not the architecture, decides whether that state beats the notes themselves. The testbed is released as a benchmark.

**Reproducibility.** Anonymized code, data generator, pre-registration, and an audio demo page (clean vs. edited continuations): <https://anonymous.4open.science/r/TBD>. **[TBD (run required): create before submission.]**

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